Customer Churn Analysis

Excel-based data consolidation, cleaning & MIS reporting on an 8,000-record telecom customer dataset

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Note: "Churn" refers to a customer discontinuing the company's service. In the dataset, the Churn field has two values — **Yes** (the customer has left / discontinued the service) and **No** (the customer is still active). Churn analysis identifies how many customers are leaving, which segments they belong to, and the resulting revenue impact.

Project Overview

The raw data was spread across three separate Excel files — customer demographics, services subscribed, and billing details — covering 8,000 customers, linked by a common CustomerID field. The objective was to consolidate this data into a single analysis-ready dataset and build MIS-style reports to understand customer churn patterns and their revenue impact, simulating the responsibilities of an MIS Executive role.

1. Data Cleaning

- Converted MonthlyCharges and TotalCharges fields from text to numeric format.
- Standardized all date fields to a single DD-MM-YYYY format.
- Applied consistent currency formatting (■) across billing fields.
- Identified missing values (blank MonthlyCharges/TotalCharges entries) for correction.

2. Data Consolidation

As the available Excel version (2007) did not support Power Query, the three files were consolidated using **VLOOKUP** formulas. A new master workbook was created with CustomerID as the base reference key. Rather than importing every available field, 13 columns most relevant to churn analysis were selected across the three source files — including SeniorCitizen, Partner, City, InternetService, OnlineSecurity, Contract, PaymentMethod, MonthlyCharges, TotalCharges and Churn.

VLOOKUP formulas were used to pull matching values from each source file based on CustomerID, applied across all 8,000 rows. To prevent broken references once the source files were closed, the formulas were converted to static values using **Paste Special** → **Values**, producing a clean, self-contained master dataset.

3. MIS Reporting (Pivot Tables)

Using the consolidated master data, Pivot Tables were built in Excel to generate churn and revenue reports, with each report placed on its own worksheet for clarity.

Contract-wise Churn Report

Contract type was placed in Row Labels, Churn status in Column Labels, and a count of CustomerID in Values. This revealed an overall churn rate of **27.80%**, fairly consistent across Month-to-Month, One Year and Two Year contracts.

C9 =2224/8000

	A	B	C	D
1				
2				
3	Total Customer	Churn		
4	Contract	No	Yes	Grand Total
5	Month-To-Month	1900	749	2649
6	One Year	1968	754	2722
7	Two Year	1908	721	2629
8	Grand Total	5776	2224	8000
9		72.20%	27.80%	

Fig 1: Contract-wise churn breakdown with overall churn rate

Revenue Report

Contract type was cross-tabulated against Churn status, summarizing total and average Monthly Charges (corrected from Excel's default "Count" aggregation to "Sum" via Value Field Settings). This quantified the revenue impact of churn at approximately **■1,49,130 in monthly recurring revenue** lost to churned customers.

Revenue Report

	churn					
	No		Yes			
Contract	Monthly Charge	Ave of Monthly Charge	Monthly Charge	Ave of Monthly Charge	Total Monthly Charge	Total Ave of Monthly Charge
Month-To-Month	131011.95	69.03	50445.81	67.35	181457.76	68.55
One Year	137166.05	69.80	51348.35	68.37	188514.4	69.41
Two Year	130135.58	68.35	47335.94	66.02	177471.52	67.71
Grand Total	**398313.58**	**69.07**	**149130.1**	**67.27**	**547443.68**	**68.57**

Fig 2: Revenue report — monthly charges by contract type and churn status

Service Report

Churn was analyzed by InternetService type (DSL, Fiber optic, No internet service), showing minimal variation in churn rate across service types for this dataset (all approximately 27-28%).

Service Report

Total customer	Churn		
Row Labels	No	Yes	Grand Total
DSL	1911	722	2633
Fiber optic	1961	754	2715
No	1904	748	2652
Grand Total	**5776**	**2224**	**8000**

Fig 3: Service report — churn by internet service type

Key Insights

Metric	Value
Total customers analyzed	8,000

Overall churn rate	27.80%
Monthly revenue lost to churn	■1,49,130
Avg. monthly charge — active customers	■69.07
Avg. monthly charge — churned customers	■67.27

Churn is distributed almost evenly across contract types, payment methods and internet service types in this dataset, rather than being concentrated in one segment as is typical in real-world telecom data. The clearest business signal is the revenue impact: the company is losing an estimated ■1,49,130 per month in recurring revenue to churned customers, reinforcing the importance of retention efforts regardless of segment.

Skills Demonstrated

Data Cleaning & Preparation	Data type correction, date standardization, missing-value identification, currency forma
Data Consolidation	VLOOKUP across workbooks, relational linking via CustomerID, column selection, Pas
Data Analysis	Pivot Tables, cross-tabulation, Value Field Settings (Sum/Average/Count), grouping, cl
MIS Reporting	Multi-report structuring, business insight generation, revenue-impact analysis
Tools	Microsoft Excel (VLOOKUP, Pivot Tables, Data Tools)

4. Power BI Dashboard

To extend the analysis beyond static Excel reports, the consolidated dataset was imported into Power BI and built into a full interactive churn dashboard, covering data modeling, Power Query transformation, DAX measures, visualization design and interactivity.

Data Import & Modeling

- Imported all 3 Excel files (demographics, services, billing) into Power BI.
- Built a data model by creating 1:1 relationships across the 3 tables via CustomerID, enabling cross-filtering across the entire model.
- Managed data types (text, number, date) to ensure calculations behaved correctly.

Power Query (Data Transformation)

- Fixed date locale/format issues during Power Query parsing.
- Re-verified cleaning done earlier in Excel (duplicates, blanks, inconsistent text) during import.

DAX (Data Analysis Expressions)

- Built core KPI measures using COUNTROWS, SUM, AVERAGE and DIVIDE (Total Customers, Total Revenue, ARPU, Avg Tenure, Avg Customer Value).
- Used CALCULATE for filtered calculations, e.g. Churn Rate (counting only "Yes" rows).
- Used SWITCH(TRUE()) to build calculated columns for segmentation (Tenure Band, Customer Segment).
- Used SELECTEDVALUE to build a dynamic measure driven by slicer selection (Monthly/Yearly Revenue toggle).
- Built a disconnected slicer-only table (RevenueView) and linked it to a DAX measure for the toggle logic.
- Applied VAR/RETURN structure for more complex, readable measures.

Data Visualization

- KPI Cards for headline metrics (Total Customers, Revenue, ARPU, Churn Rate, Avg Tenure, Avg Customer Value).
- Clustered bar/column charts for comparisons (Contract-wise Churn, Service Adoption).
- Donut charts for distribution (Internet Service mix, Payment Method mix).
- Combo chart (line + column) to compare Revenue and Churn Rate together over months.
- Scatter chart for multi-dimensional analysis (Customer Value vs Churn, bubble size = count).
- Table visual for a detailed demographic breakdown.

Formatting, Design & Interactivity

- Custom dark theme applied consistently across the dashboard using precise hex color codes.
- Conditional/category-based coloring via the fx function, and gradient + data-bar formatting inside the table visual.
- Custom data-label formatting (% symbols, decimal precision).
- Multiple slicers (City, Contract Type, Internet Service, Payment Method, Tenure Band) with full cross-filtering across visuals.

Business Analysis Thinking

- Selected business-meaningful KPIs beyond basic churn % — including ARPU and Average Customer Value.
- Built segmentation logic to group customers meaningfully (High/Medium/Low value, Tenure bands).
- Wrote data-driven Key Insights directly from chart patterns, avoiding unsupported conclusions.

Note: All data cleaning, consolidation, Pivot Table reporting, and the Power BI dashboard (data modeling, Power Query, DAX, visuals) were built hands-on in Excel and Power BI. AI assistance (Claude) was used for guidance, troubleshooting, and structuring this project write-up.